

# Coefficient of Variation

```
pacman::p_load(ggplot2, #plots
               tictoc, #timing
               dplyr, #filter
               e1071 #skewness
               )
```

```
source("../functions_main/calculate_trait&calculate_demo_v10.R")
```

## 500 simulations and CV analysis

I computed separately on the server 500 simulation for each setup:

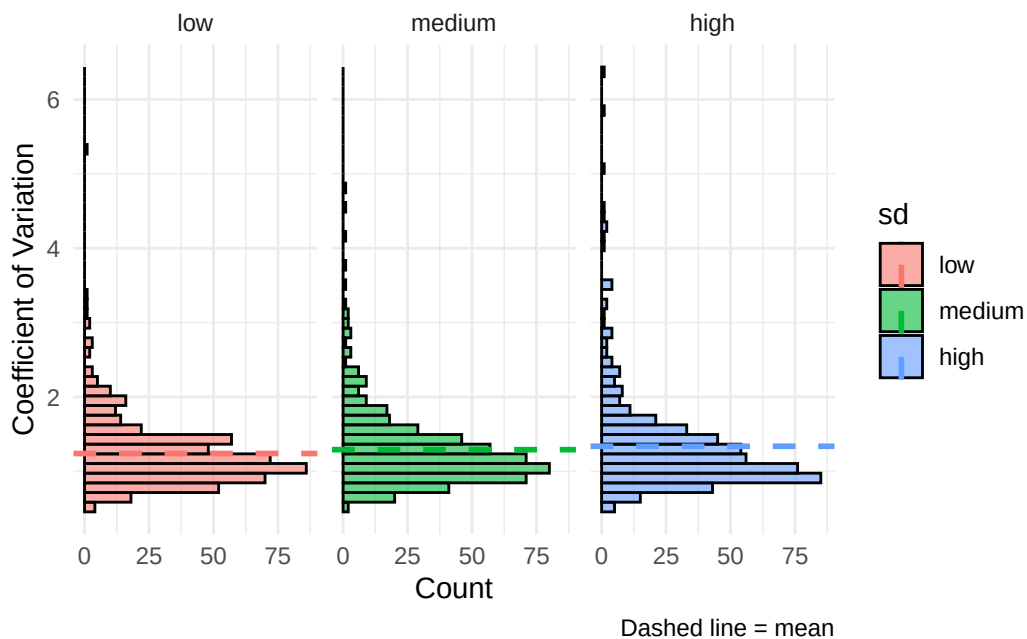
| Coefficient | Trait | Type     | Standard deviation | Trait |
|-------------|-------|----------|--------------------|-------|
| 0.55        |       | Small    | 0.05               |       |
| 0.55        |       | Moderate | 0.1                |       |
| 0.55        |       | High     | 0.15               |       |

this is the result with much more detail:

```
small<-readRDS("../server_runs/run_sim/RDS/final_small_sd.rds")
moderate<- readRDS("../server_runs/run_sim/RDS/final_moderate_sd.rds")
high <- readRDS("../server_runs/run_sim/RDS/final_high_sd.rds")
```

```
cv_combined_500 <- bind_rows(
  data.frame(cv = coefficient_of_variation(small), sd = "low"),
  data.frame(cv = coefficient_of_variation(moderate), sd = "medium"),
  data.frame(cv = coefficient_of_variation(high), sd = "high")) |>
  mutate(sd = factor(sd, levels = c("low", "medium", "high")))
```

```
ggplot(cv_combined_500, aes(x = cv, fill = sd)) +
  geom_histogram(binwidth = 0.13, color = "black", alpha = 0.6) +
  geom_vline(data = cv_combined_500 %>%
    group_by(sd) %>%
    summarise(mean_cv = mean(cv)),
    aes(xintercept = mean_cv, color = sd),
    linetype = "dashed", linewidth = 1) +
  labs(x = "Coefficient of Variation", y = "Count", caption = "Dashed line = mean") +
  theme_minimal() +
  coord_flip() +
  facet_wrap(~sd, ncol = 3)
```



```
cv_combined_500 |>
  group_by(sd) |>
  summarise(mean_cv = mean(cv))
```

```
# A tibble: 3 x 2
  sd      mean_cv
<fct>   <dbl>
1 low      1.24
2 medium   1.29
3 high     1.34
```

We see that the CV increases linearly as the standard deviation of the trait linearly increases (0.05, 0.10, 0.15)

```
cv_combined_500 |>
  group_by(sd) |>
  summarise(median_cv = median(cv))
```

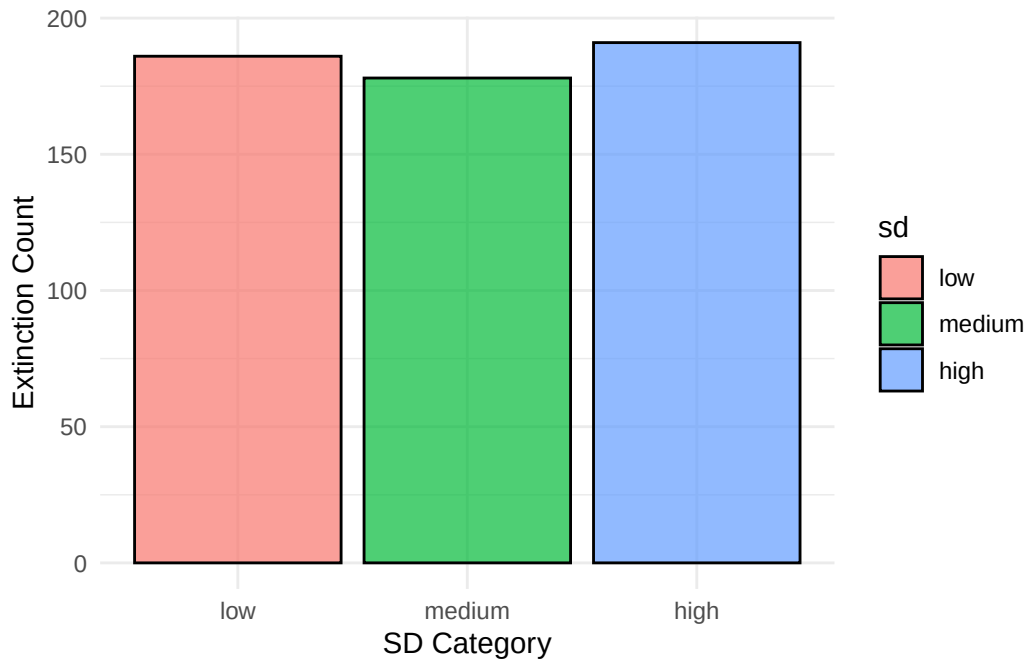
```
# A tibble: 3 x 2
  sd      median_cv
<fct>      <dbl>
1 low        1.13
2 medium     1.17
3 high       1.16
```

If we chose the median statistic the difference between the standard deviation are even smaller. The median is robust to skewness.

```
extinction_500 <- c(extinct_count(small), extinct_count(moderate), extinct_count(high))
sd_labels <- c("low", "medium", "high")

extinction_df_500 <- data.frame(
  count = extinction_500,
  sd = factor(sd_labels, levels = sd_labels)
)

ggplot(extinction_df_500, aes(x = sd, y = count, fill = sd)) +
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  aes(color=sd)+
  labs(x = "SD Category", y = "Extinction Count") +
  theme_minimal()
```



With 500 simulations, we now have enough resolution. Medium standard deviation shows the lowest number of extinctions, followed by low and then high standard deviation.

### When they go extinct?

We now check the distribution of the year in which the population go extinct, I expect to have a sort of uniform distribution.

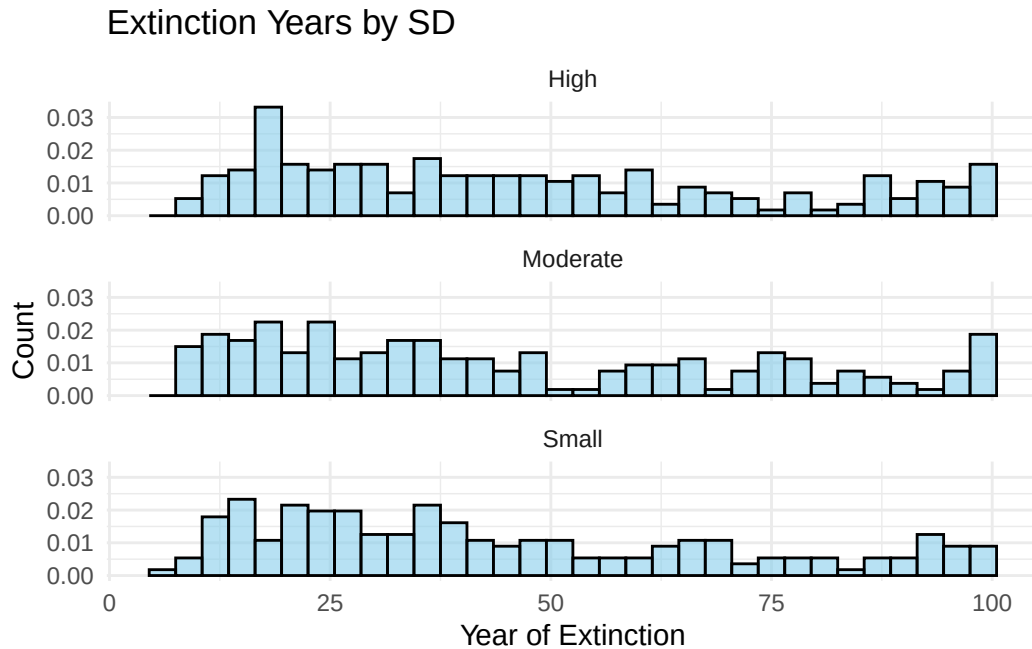
```
extinct_year <- function(table){
  table |>
    group_by(simulation) |>
    filter(n() < 101)|>
    summarise(extinct_year=max(year)) |>
    pull(extinct_year)
}
```

```
year_small <- data.frame(year = extinct_year(small), sd = "Small")
year_moderate <- data.frame(year = extinct_year(moderate), sd = "Moderate")
year_high <- data.frame(year = extinct_year(high), sd = "High")

year_all <- rbind(year_small, year_moderate, year_high)

ggplot(year_all, aes(x = year)) +
```

```
geom_histogram(aes(y = after_stat(density)), binwidth = 3, color = "black", alpha = 0.6, f
facet_wrap(~sd, ncol = 1, scales = "fixed") +
theme_minimal() +
labs(x = "Year of Extinction", y = "Count", title = "Extinction Years by SD")
```



The distribution appears left-skewed, with extinctions occurring a little more within the first 50 years of the simulation. This skew is likely due to the low initial population size (200 individuals). After a few years, the number of surviving individuals becomes more variable (ranging from 1 to around 20,000 in most cases), and the population's expected time to extinction becomes roughly uniform.

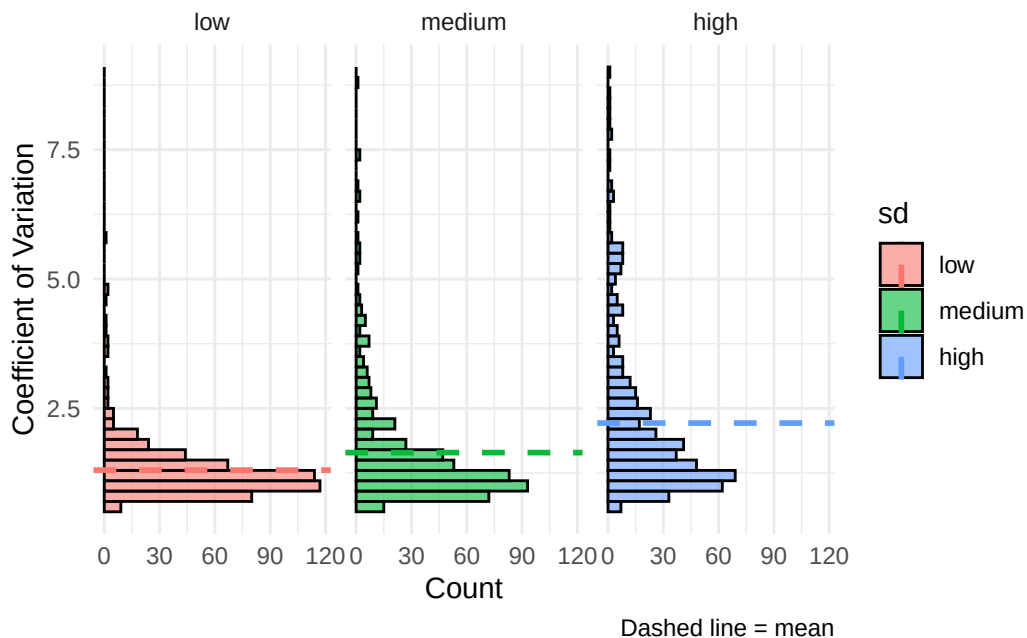
#### Analysis with a different parameter table (double of s.d.)

| Coefficient | Trait | Type      | Standard deviation | Trait |
|-------------|-------|-----------|--------------------|-------|
| 0.55        |       | Small2    | 0.1                |       |
| 0.55        |       | Moderate2 | 0.2                |       |
| 0.55        |       | High2     | 0.3                |       |

```
small2<- readRDS("../server_runs/run_sim2/RDS2/final_small_sd_2.rds")
moderate2<- readRDS("../server_runs/run_sim2/RDS2/final_moderate_sd_2.rds")
high2<- readRDS("../server_runs/run_sim2/RDS2/final_high_sd_2.rds")
```

```
cv_combined_500_2 <- bind_rows(
  data.frame(cv = coefficient_of_variation(small2), sd = "low"),
  data.frame(cv = coefficient_of_variation(moderate2), sd = "medium"),
  data.frame(cv = coefficient_of_variation(high2), sd = "high")) |>
  mutate(sd = factor(sd, levels = c("low", "medium", "high")))
```

```
ggplot(cv_combined_500_2, aes(x = cv, fill = sd)) +
  geom_histogram(binwidth = 0.20, color = "black", alpha = 0.6) +
  geom_vline(data = cv_combined_500_2 %>%
    group_by(sd) %>%
    summarise(mean_cv = mean(cv)),
    aes(xintercept = mean_cv, color = sd),
    linetype = "dashed", linewidth = 1) +
  labs(x = "Coefficient of Variation", y = "Count", caption = "Dashed line = mean") +
  theme_minimal() +
  coord_flip() +
  facet_wrap(~sd, ncol = 3)
```



What is clear from this plot is an increase on the skewness of the distribution when choosing higher level of standard deviations.

```
cv_combined_500_2 |>
  group_by(sd) |>
  summarise(mean_cv = mean(cv))
```

```
# A tibble: 3 x 2
  sd      mean_cv
<fct>    <dbl>
1 low      1.31
2 medium   1.64
3 high     2.22
```

```
cv_combined_500_2 |>
  group_by(sd) |>
  summarise(median_cv = median(cv))
```

```
# A tibble: 3 x 2
  sd      median_cv
<fct>    <dbl>
1 low      1.17
2 medium   1.27
3 high     1.66
```

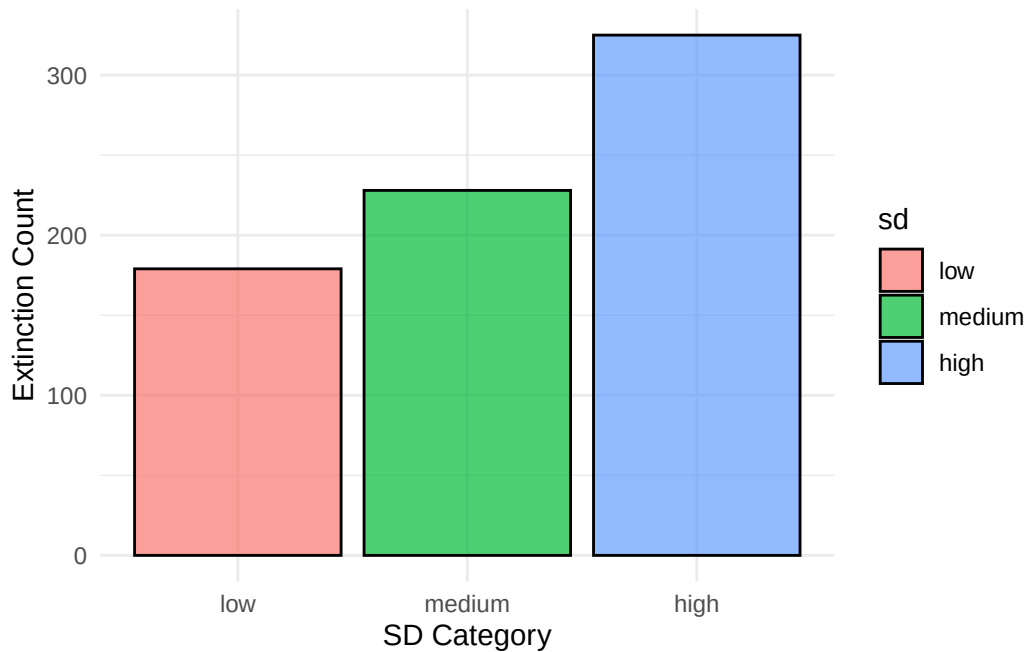
```
cv_combined_500_2 |>
  group_by(sd) |>
  summarise(skewness = skewness(cv))
```

```
# A tibble: 3 x 2
  sd      skewness
<fct>    <dbl>
1 low      3.29
2 medium   2.66
3 high     1.78
```

```
extinction_500_2 <- c(extinct_count(small12), extinct_count(moderate2), extinct_count(high2))
sd_labels <- c("low", "medium", "high")

extinction_df_500_2 <- data.frame(
  count = extinction_500_2,
  sd = factor(sd_labels, levels = sd_labels)
)
```

```
ggplot(extinction_df_500_2, aes(x = sd, y = count, fill = sd)) +
  geom_bar(stat = "identity", color = "black", alpha = 0.7) +
  aes(color=sd)+
  labs(x = "SD Category", y = "Extinction Count") +
  theme_minimal()
```



```
year_small2 <- data.frame(year = extinct_year(small2), sd = "Small")
year_moderate2 <- data.frame(year = extinct_year(moderate2), sd = "Moderate")
year_high2 <- data.frame(year = extinct_year(high2), sd = "High")

year_all2 <- rbind(year_small2, year_moderate2, year_high2)

ggplot(year_all2, aes(x = year)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 3, color = "black", alpha = 0.6, fill = "white") +
  facet_wrap(~sd, ncol = 1, scales = "fixed") +
  theme_minimal() +
  labs(x = "Year of Extinction", y = "Count", title = "Extinction Years by SD")
```



## Extinction Years by SD

